

**INSTITUTE OF ENGINEERING & MANAGEMENT  
NEWTOWN, KOLKATA**

**Department of Electronics & Communication  
Engineering**

**Module 6 – IVI In-Vehicle Infotainment Systems Project  
Conceptual IVI Architecture with Web-Based Prototype**

**Submitted by: PARTHIBA SARKAR**

Enrollment Number: 12023002002106

Registration Number: 304202300900493

Class Roll Number: B-33

## 1. Abstract

This project presents a conceptual In-Vehicle Infotainment (IVI) architecture that integrates media playback, navigation, phone connectivity and projection/HUD functions through a common system. A lightweight browser-based prototype was developed using HTML, CSS and JavaScript to demonstrate the main IVI screens and basic interactions. The conceptual architecture separates user and vehicle inputs, IVI applications, core services and output interfaces. The prototype provides Home, Media, Navigation, Phone and Projection screens and demonstrates how information can flow between these functions.

## 2. Introduction

An In-Vehicle Infotainment system is an electronic platform that combines information and entertainment services inside a vehicle. Modern IVI systems may receive inputs from a touchscreen, steering controls, voice interface, GPS receiver and connected smartphones. These inputs are processed by application services and supporting middleware before information is delivered through the IVI display, vehicle speakers, projection display or other vehicle interfaces.

The objective of this project is to create a conceptual IVI architecture and demonstrate it through a functional web-based prototype. The prototype is intended as an educational demonstration of IVI interfaces and service interactions rather than a production automotive system.

## 3. Problem Statement

An IVI system needs to coordinate several user-facing functions such as media, navigation, phone connectivity and information projection. A clear architecture is required so that these functions can communicate through well-defined services while providing understandable outputs to the driver. This project addresses the requirement by designing a conceptual layered architecture and implementing a simple interactive browser prototype.

## 4. Objectives

- Design a conceptual architecture for an In-Vehicle Infotainment system.
- Integrate media, navigation, phone connectivity and projection functions.
- Describe the interfaces between user/vehicle inputs, IVI applications and core services.
- Develop a lightweight web-based prototype for demonstrating the IVI concept.
- Demonstrate basic interactions such as media play/pause, navigation display and call state.
- Show how navigation information can be presented on both the main IVI interface and a HUD-style projection interface.
- Document the working and test observations of the prototype.

## 5. Conceptual IVI Architecture

The proposed architecture is organized into four main layers: Inputs, IVI Applications, Core Services/Middleware, and Outputs/Vehicle Interfaces. The overall data flow is:

**User / Vehicle Input → IVI Applications → Core Services / Middleware → Display / Audio / Projection Output**

| Inputs            | IVI Applications | Core Services / Middleware | Outputs / Interfaces     |
|-------------------|------------------|----------------------------|--------------------------|
| Touchscreen       | Media Player     | Media Service              | IVI Display              |
| Steering Controls | Navigation App   | Navigation Service         | Vehicle Speakers         |
| Voice Input       | Phone / Call UI  | Connectivity Service       | HUD / Projection Display |
| GPS Receiver      | Projection UI    | Audio/Video Manager        | Vehicle Network          |
| Bluetooth / Phone |                  | Projection Service         |                          |

The architecture separates the application layer from supporting services. For example, the Navigation App can obtain position information through the Navigation Service, while the Projection Service can send selected navigation guidance to a HUD-style output.

## 6. Component Interfaces

| Interface                                          | Function                                                                            |
|----------------------------------------------------|-------------------------------------------------------------------------------------|
| Touchscreen / Steering Controls → IVI Applications | Provides commands such as play, pause, route selection and menu navigation.         |
| GPS Receiver → Navigation Service                  | Provides positioning information used by the navigation function.                   |
| Bluetooth / Phone → Connectivity Service           | Provides smartphone, call and media connectivity.                                   |
| Media Service → Audio/Video Manager                | Transfers selected media and playback information for processing.                   |
| Navigation Service → Navigation App                | Provides route, position and guidance information.                                  |
| Projection Service → HUD / Projection Display      | Sends relevant information such as navigation guidance to the projection interface. |
| Audio/Video Manager → Speakers / Display           | Delivers processed audio and visual output.                                         |
| Vehicle Network → IVI System                       | Provides selected vehicle-related information to the infotainment system.           |

## 7. Prototype Implementation

A lightweight web-based prototype was created using HTML, CSS and JavaScript. The project folder is named Wipro\_IVI\_System\_Project and the main file is index.html. The prototype uses a single browser interface with separate functional screens for Home, Media, Navigation, Phone and Projection.

The Home screen provides the main IVI dashboard. The Media screen demonstrates playback controls. The Navigation screen presents a conceptual GPS and route interface. The Projection screen presents navigation guidance in a HUD-style layout. The Phone screen demonstrates smartphone connectivity and a basic call-state interaction.

## 8. Practical Demonstration Screenshots

The following screenshots document the working prototype and provide evidence of the implemented web-based interface.

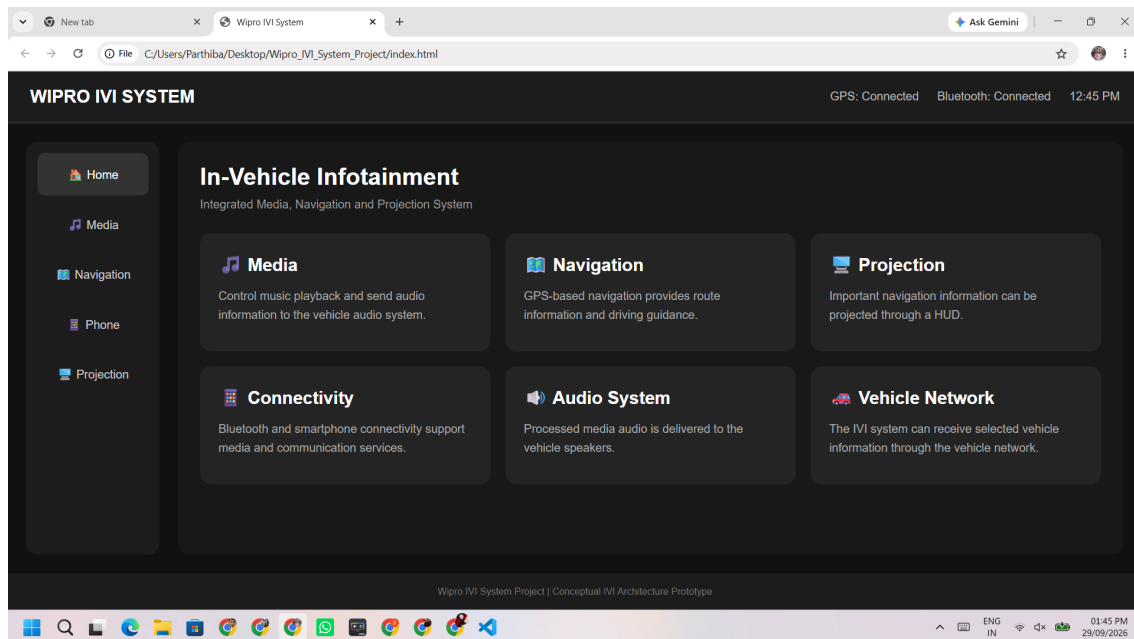


Figure 1. IVI Home Dashboard

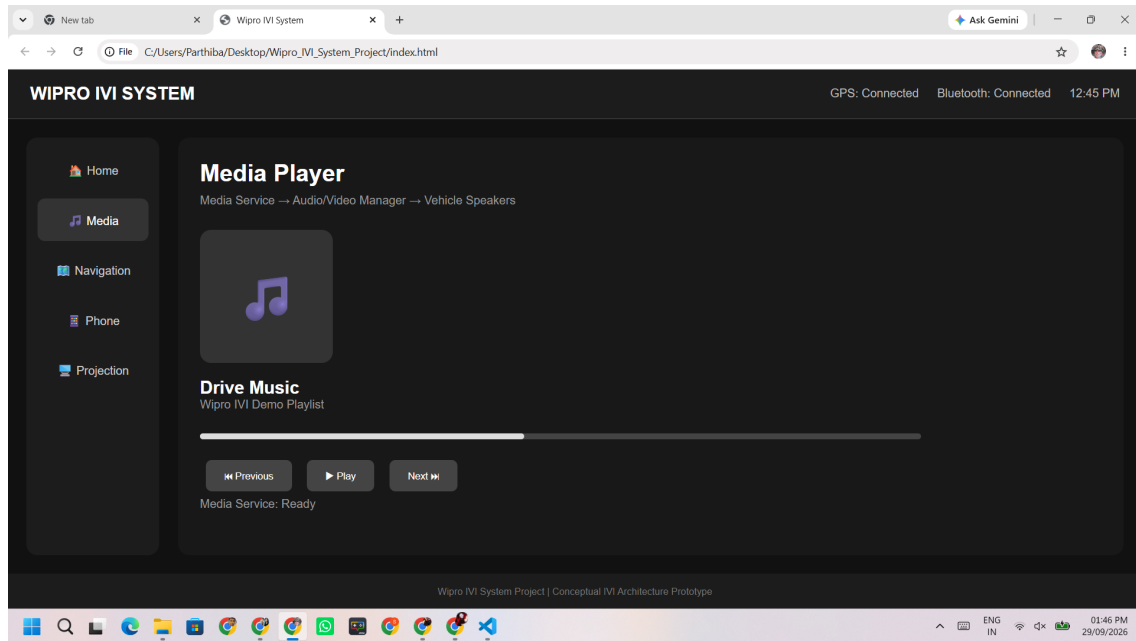


Figure 2. Media Player Interface

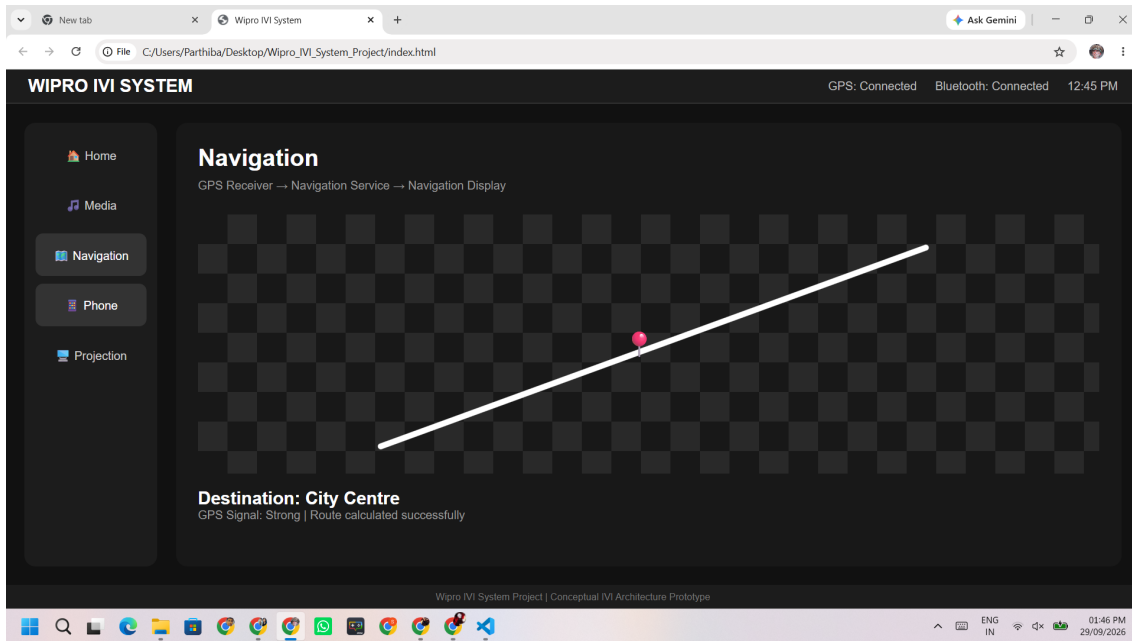


Figure 3. Navigation Interface

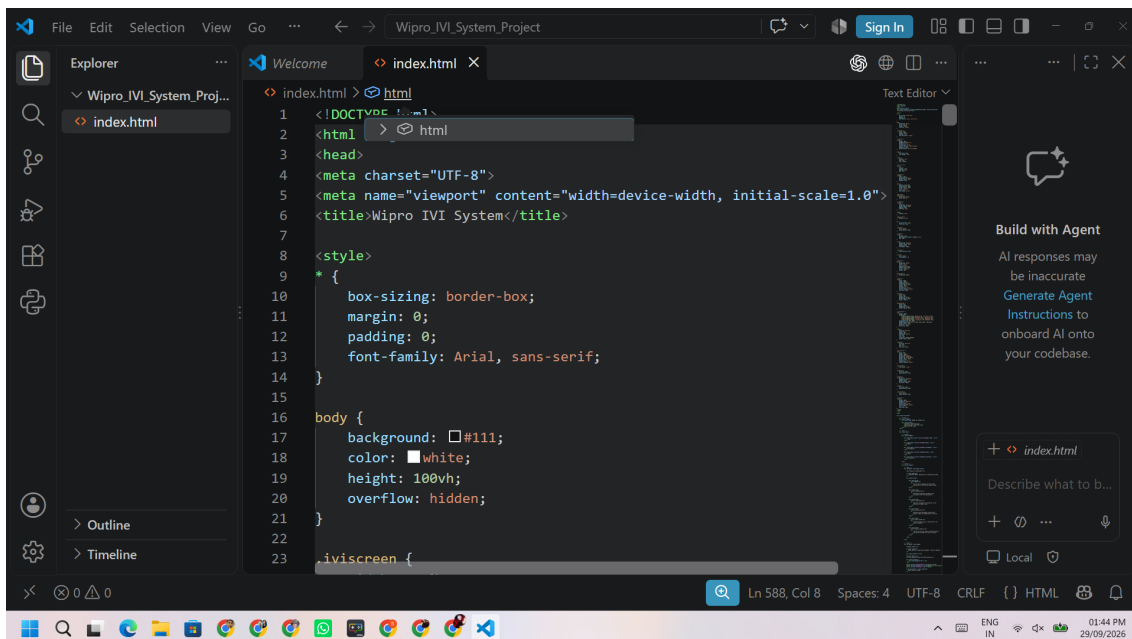


Figure 4. Projection / HUD Interface

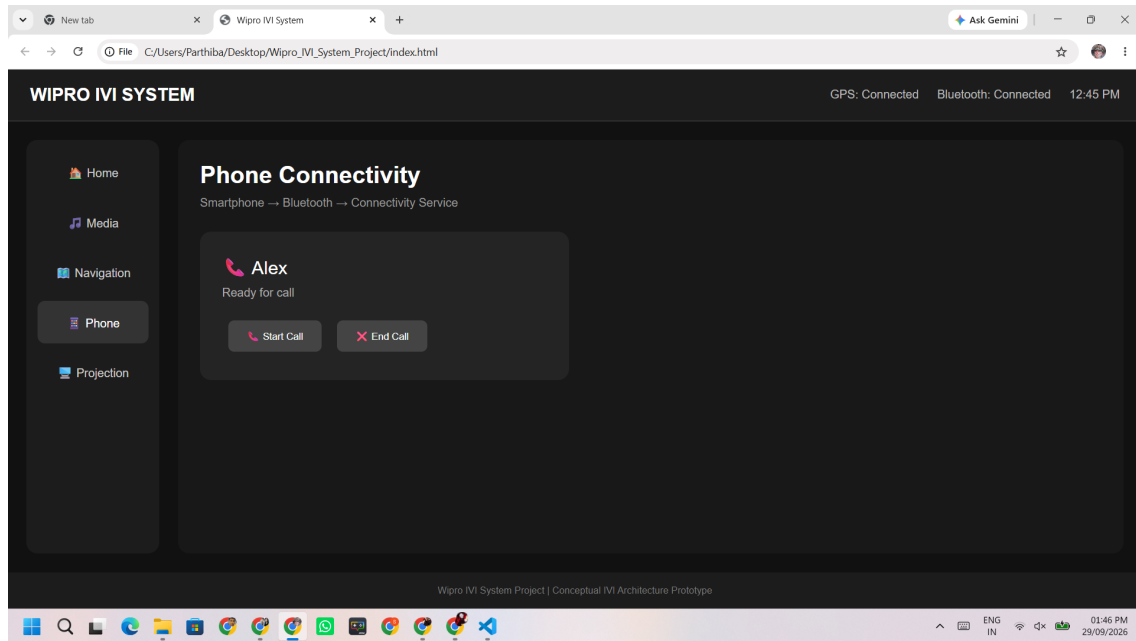


Figure 5. Phone Connectivity Interface

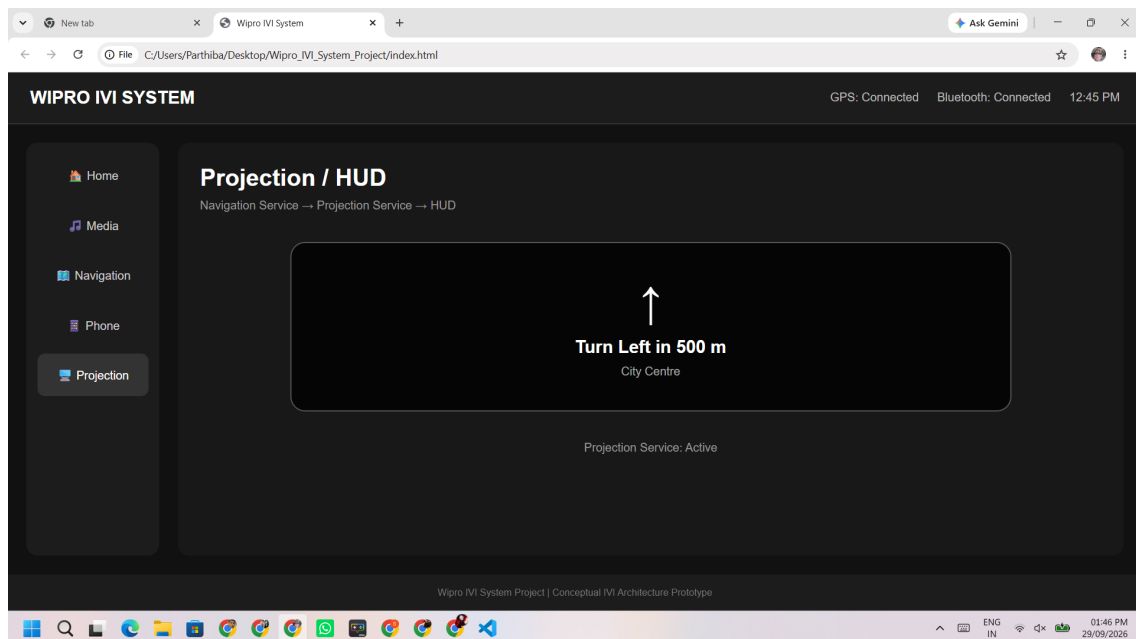


Figure 6. HTML Source Code in Visual Studio Code

## 9. Working of the Prototype

- The Home screen presents the integrated IVI functions.
- The Media screen demonstrates media service interaction and play/pause control.
- The Navigation screen demonstrates GPS-to-navigation data flow using a route visualization.
- The Projection screen demonstrates navigation guidance being presented through a HUD-style display.
- The Phone screen demonstrates smartphone connectivity and call-state interaction.
- The Navigation and Projection interfaces show how the same conceptual navigation information can be presented through different outputs.

## 10. Test Results

| Test           | Action                          | Observed Result                                                |
|----------------|---------------------------------|----------------------------------------------------------------|
| Home Dashboard | Open index.html                 | IVI dashboard loads successfully.                              |
| Media          | Open Media and press Play       | Media status changes to playing.                               |
| Navigation     | Open Navigation                 | Route and GPS status are displayed.                            |
| Projection     | Open Projection                 | HUD guidance is displayed and Projection Service shows active. |
| Phone          | Open Phone and press Start Call | Call status changes to connected.                              |

## 11. Results and Discussion

The completed prototype demonstrates the main conceptual IVI services through an interactive browser interface. The project shows how media, navigation, phone connectivity and projection can be represented as separate applications while sharing common core services. The practical demonstration confirms that the main screens load and that the implemented controls update the corresponding interface states.

The prototype is intentionally lightweight and conceptual. It does not communicate with actual vehicle hardware, GPS hardware, Bluetooth modules, CAN buses or a real HUD. Therefore, the demonstrated results represent software-level interaction rather than real automotive integration.

## 12. Conclusion

This project implements a conceptual IVI architecture and a functional web-based prototype for demonstrating its main interfaces. The architecture organizes user and vehicle inputs, IVI applications, core services and output interfaces into a clear data-flow structure. The browser prototype provides practical demonstrations of Home, Media, Navigation, Phone and Projection functions, providing a foundation for further development of a more realistic automotive infotainment system.

## 13. Future Scope

- Integration with real GPS and vehicle sensor inputs.
- Bluetooth-based smartphone connectivity and real call/media services.
- CAN or automotive Ethernet integration for vehicle information.
- Integration with a real head-up display or projection hardware.
- Voice-command support for hands-free IVI interaction.
- Development of a responsive mobile or embedded IVI interface.
- Integration with real navigation map services and live route data.
- Deployment on an automotive embedded platform.

## 14. Tools and Technologies

- HTML5
- CSS3
- JavaScript

- Visual Studio Code
- Web Browser

## 15. Project Files

- Wipro\_IVI\_System\_Project – project folder.
- index.html – main web-based IVI prototype.
- Wipro\_IVI\_System\_Project\_Report\_With\_Screenshots – project report.

## 16. References

- MDN Web Docs — HTML, CSS and JavaScript documentation.
- W3C Web Standards — HTML and CSS specifications.
- General IVI system architecture concepts used for the educational prototype.